

Micro I, class 9

I initiated a discussion of another way to do comparative statics. I started from a classical case: $f : \mathbb{R}^2 \rightarrow \mathbb{R}$ has two real numbers as arguments, x , a choice variable, and t , a parameter. Assume that f is C^2 and satisfies

$$f_x(x, t) = 0 \Rightarrow f_{xx}(x, t) < 0$$

for every $t \in \mathbb{R}$. This is called *strong quasiconcavity* for functions on the real line (it implies strict quasiconcavity).¹ Assume too that for every $t \in \mathbb{R}$, there is a solution $x^*(t)$ to the problem of picking $x \in \mathbb{R}$ to maximize $f(x, t)$. By strong quasiconcavity, $x^*(t)$ is single-valued. It must satisfy the necessary first-order condition

$$f_x(x^*(t), t) = 0$$

for every $t \in \mathbb{R}$. Since $f_{xx}(x^*(t), t) < 0$ the *implicit function theorem* (or simple implicit differentiation) implies that

$$x^{*'}(t) = -\frac{f_{xt}(x^*(t), t)}{f_{xx}(x^*(t), t)}.$$

Since $f_{xx}(x^*(t), t) < 0$ (by strong quasiconcavity), it follows that $x^{*'}(t) > 0 \ \forall t \in \mathbb{R}$ iff $f_{xt}(x^*(t), t) > 0 \ \forall t \in \mathbb{R}$. The last inequality is equivalent in this setting to “ $f_x = 0 \Rightarrow f_{xt} > 0$.” And the last implies that f_x , viewed as a function of t , *crosses zero at most once from below*.

I used this classical result as inspiration for the *strict single crossing property* (SSCP). It asserts that, for any $x'' > x'$, the difference $f(x'', t) - f(x', t)$, viewed as a function of t , crosses zero at most once from below. I used this definition to prove a very special case of a comparative statics theorem attributed to Milgrom and Shannon (1994).² This is the Theorem on page 1 of my notes. I hope I convinced you about the application of the theorem to applied MWG, Problem 3.G.4, at least for the case of two goods (Exercise 5 on page 1 in my notes on comparative statics).

I strongly recommend that you download the Milgrom-Shannon paper. Certainly not to read line-by-line — it goes *far* beyond what we will do (though I can't *ban* you from reading it all) — but for the examples, most of which can be understood using the very special case of their results that I am presenting. I already mentioned in my last summary works by Vives, Sundaram, and Amir-Rietzke, whose examples and exercises you might find helpful.

¹See Diewert, W. Erwin, Mordecai Avriel, and Israel Zang. “Nine kinds of quasiconcavity and concavity.” *Journal of Economic Theory* 25, no. 3 (1981): 397-420.

²Theorem 4' in Milgrom, Paul, and Christina Shannon. “Monotone comparative statics.” *Econometrica* (1994): 157-180.